SAR HIGH SCHOOL 2023-2024

ROBOSAPIENS

Team 19167



TABLE OF CONTENTS

Our Team 2

Overall Plan 3

Meet Our Robot! 4-5

Design Evolution 6-8

Game Strategy 9

Outreach 10-14

Obstacles 15

I: OUR TEAM

COACHES: Jonathan Martin, Felix Caso

CO-CAPTAINS:
Ilan G. (12) Captain
Kerenna K. (12) Captain

SQUAD MEMBERS:
Jamie S. (12)
Jeremy B. (12)
Barak W. (12)
Emerson K. (12)
Eytan R. (11)
Asher k. (11)
Hila L. (10)
Ilay G. (10)
Tamar R. (10)

We are *RoboSapiens*, FTC Team 19167, a squad composed of students from SAR High School in Riverdale, New York. Our School's String Robotics Team consists of 32 members and our squad has 10 members. This is our sixth year of participating in the FIRST Tech Challenge (FTC) competition. We are all united as a team with our love for science, technology, engineering, and mathematics!



II: OVERALL PLAN

Season Goal:

This year, we put extra effort into building the best robot we can. We increased practice time from once to twice a week, organized a dynamic team structure for maximum creativity and efficiency, and constantly tested and upgraded our robot. The goal of our team is not only to do well in the FTC competition but also to build a community that fosters and spreads the love of engineering. We also made it a team goal to improve our networking and outreach skills for next year by developing closer connections with other teams earlier on.

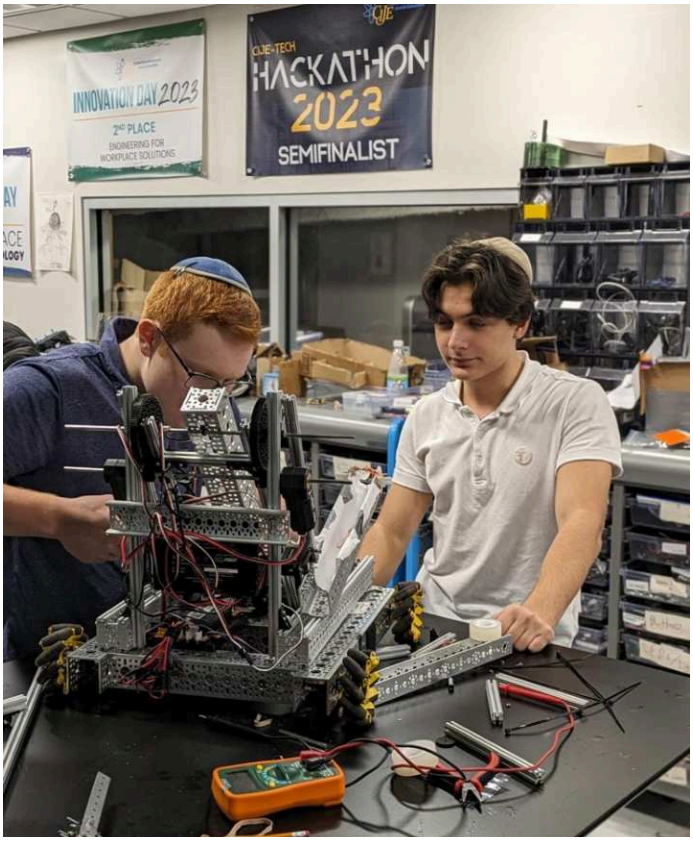
Our Team Dynamic:

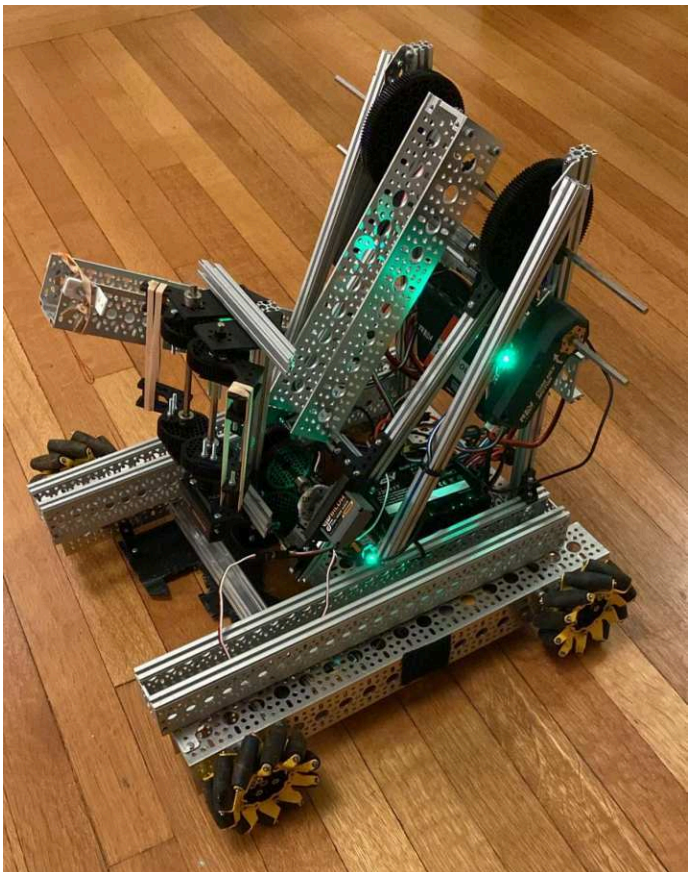
Our team puts a special emphasis on opting in and being excited about robotics. All of our members are committed and engaged, and with each of our different skill sets, we collaborate to achieve a shared objective. We began the building process by making sure to take our time to brainstorm different ideas and encouraging everyone to pitch their designs. We then worked together to construct prototypes and evaluate each option's efficiency.

Our Strategies for Designing and Learning:

Once we decided on a design, our designated team members went to work on constructing 3D models using Fusion 360 to give us a mapped-out diagram of the parts. Other team members followed online tutorials and helped each other to learn Fusion and improve their digital designing skills. We then got to work building the robot, making sure to delegate different jobs to team members to prevent crowding and get our first working robot finished in only a few weeks. However, our learning doesn't stop there, as we make sure to test out each aspect of our robot and evaluate how it can be improved. The next part of our plan is to develop our autonomous robot and practice our driving skills to make sure we're prepared for the competition!

III: MEET OUR ROBOT!





Meet our robot, Picasso! Our robot's name was inspired by its artistic abilities as you will see later in the Outreach chapter. Picasso not only loves to paint in CenterStage by placing pixels on an easel to create beautiful mosaics but is also skilled in using its adaptable arm to hold brushes to paint!

Piece By Piece:

Chassis - The rectangular chassis provides a strong foundation to support the rest of our robot. We implemented a larger and sturdy chassis so that our robot could speedily drive through the field, and used omnidirectional mecanum wheels for ultimate speed and precision in our driving.

Arm - We use an arm with one joint at the top. The joint is powered by a gear connected to a hex motor. The arm moves up and down and goes all the way around the top to place pixels effectively.

Intake - We have a four-pronged claw system to grab pixels. It has a servo-powered wrist to allow easy access to pick up the pixels. We also have a cardboard scooper piece at the bottom of our robot that adjusts the pixels into an optimal position for the claw to grab them. The robot drives into the pixels and the unique shape of the scooper forces the pixels into the desired position.

Launcher - The paper airplane launcher is a long metal pathway meant to fit our drone. A rubber band is pulled back onto a servo motor collecting mass amounts of potential energy. The drone is then fitted in between the rubber band so that when the servo releases, as per our instruction to do so, the launch occurs, propelling our drone to its desired drop point.

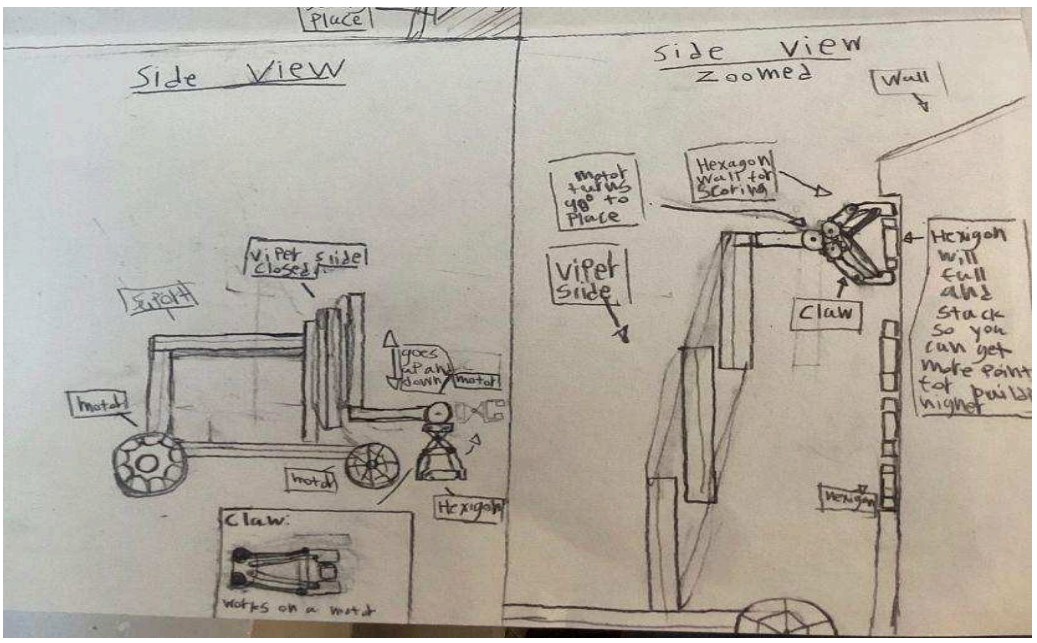
Coding - Our TeleOp software was designed to ensure ease of use for the robot driver. Having the ability to both turn and strafe ensures the robot can easily navigate through complex obstacles. For Autonomous mode, our robot utilizes our motors' built in encoders to move precisely. We take advantage of our robot webcam to use computer vision to recognize our team's custom prop.

IV: DESIGN EVOLUTION

Intake evolution:

For the claw intake, our goal was to create a mechanism not only safe and reliable but also swift and time-efficient. The initial concept picked up a pixel while another one would be stored below allowing us to almost double our speed. At first, a viper slide was considered as a way to elevate the pixels but this idea was upgraded into an arm that places the pixels behind where it picks them up as it enhances our reach into the board, ensuring all pixels land in the correct spot. In refining our design, we drew inspiration from the arm mechanism employed in our previous year's robot, making necessary adjustments and fine-tuning it to align with our objectives. We soon realized that the claw had issues grabbing pixels that weren't positioned in a certain way, so we had the idea to add a scooper piece at the bottom of our robot. As our robot drives toward the pixel, the unique shape of the scooper adjusts the pixel into a position that allows it to be easily grabbed by our claw. This decision proved to be remarkably time-efficient, as the simplicity of the idea facilitated smooth execution, bringing us one step closer to achieving our goal.

- *Ilay Gibor and his initial sketch below*

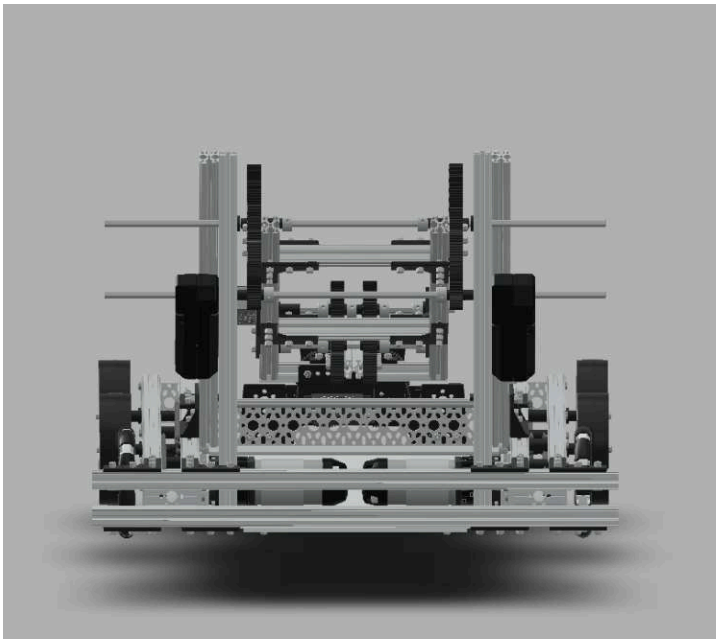


3D Modeling - Fusion 360:

I decided to learn Fusion 360 to help digitally design our robot. I used YouTube tutorials, help from my teammates, and trial and error to build this complete CAD of our robot. The first step for me in this design process was to brainstorm all the different aspects our robot would need, and play around with figuring out how they would interact with each other. I experimented with how positioning parts on the robot would affect its efficiency, without having to physically build each idea, allowing me a much better way to narrow down design options. Then, I sketched out the structure of how I thought the robot would come together, learning how to create sketches on Fusino 360 and extrude them to create different shapes. Finally, I was left with a fully digitalized model of our robot, and a lot more valuable knowledge on how to use CAD. This model provided our team with a framework to experiment virtually without risking damage to the robot, led to new ideas for our design evolution, and allowed multiple people to work on the robot at the same time virtually, so that way everyone can learn and help without it getting too crowded.

- *Asher Kaynan and his CAD below*





Part of our team's strategy was to use our first qualifier match to alert us to the main issues that would arise during gameplay and then solve them before our next competition. Here are some of the main design issues we had and how we fixed them:

- **Drone Launcher Placement**

We realized that the drone launcher's spot on the side of our robot prevented us from being able to drive the robot under the truss because the preloaded paper airplane would then be knocked off. As a result, our drone feature didn't work during the first qualifier. We experimented with other places to put the launcher, such as the top of the arm, but that also interfered with the truss. We then placed the drone launcher on the bottom of the arm, which worked really well because it was protected from the truss while passing through and we were also able to adjust the arm's angle when shooting the launcher to get the perfect trajectory for our airplane.

- **A sticky situation**

Our team did not consider the huge effect of our tape choice on our robot's intake before the first qualifier. Our scooper piece on the bottom of the robot has tape lining it to help grab pixels easier, however, the tape was so sticky that during our matches the pixels kept getting stuck and not lifting during autonomous! We managed to fix this problem because of the help of other FTC teams, as we borrowed and then tested many different kinds of their tape to see which worked best and solved our issue.

- **Servo Switcharoo**

Our servo piece was the biggest issue in our matches, frequently preventing our intake from moving at all. After careful inspection, we realized that our 3D-printed servo piece had been stripped from the pressure of the wrist joint. We replaced this piece with a stronger metal material to strengthen our intake and allow it to last longer.

V: GAME STRATEGY

- ➔ Our game strategy focused on trying to maximize our time to score points. One way we accomplished this was through our double intake of pixels. With one form of intake being the claw on our robot and the other being the cardboard pixel scooper we not only increase our intake security but also our robot to gather pixels quicker. By having our robot's arm able to swing backward to place pixels on the easel, we also save time because we don't have to turn our robot around before each placement.
- ➔ Another factor we prioritized in our robot was speed. We added a metal ramp to the base of our arm to automatically allow our robot to easily lift the gates on the field and drive under. Our omnidirectional wheels and sleek chassis also contribute to our ability to quickly navigate the field.
- ➔ We carefully chose which game aspects to focus on in order to maximize our playing time. Our reliable intake allowed our team to strongly utilize autonomous mode, and our team spent a lot of time improving the code to score as many points as we could. We focused on quicker game elements such as the drone launcher and didn't include more time consuming ones such as the lift mechanism so that we could maximize our gameplay time to score more points through pixel placement.

Our team believes that an important part of a team strategy is keeping up team morale and energy. Team member Eitan helped lift everyone's spirits by baking waffles in our design lab during our practices!



VI: OUTREACH

iShine Trip:

Our Robotics team is committed to spreading the excitement and education of engineering. We took a group trip to **IShine**, a program designed for children living in homes that have been affected by illness or loss in their families, to introduce the kids there to the world of STEM. We brought all different types of children's engineering kits and built a bunch of fun projects with the kids while teaching them about different engineering tactics and tools. We also made sure to include more art-centered building kits to engage students of all ages and interests. From bottle rocket launchers, motor-powered motorcycles, conveyor belts, portable projector movies, and so much more, the children were all incredibly amazed at what their creations could do.

After the kids were thrilled about their own mini-builds, we brought out our robot, which blew them all away. They had so much fun driving the robot and playing around with the drone launcher. The children each colored a piece of paper and learned to fold it into a paper airplane, which they were then able to place in the drone launcher and press the button to send it flying! They were so interested in how our robot worked, and we really felt as if we had spread the contagious love of engineering. Our team left iShine feeling so proud and fulfilled at the knowledge we had imparted, and even more importantly, the amazing smiles and genuine excitement that we brought to these kids.

An excerpt from an email our principal received after our visit from an onlooking volunteer:

"This afternoon the SAR Robotics Club came to Westchester Day School to work with the children in the I-Shine after-school program. Together they created and built amazing robotic items. They kept all the children engaged and excited. Your SAR students were kind, patient and respectful! It was a fun afternoon!"





Spuyten Duyvil Library Event:

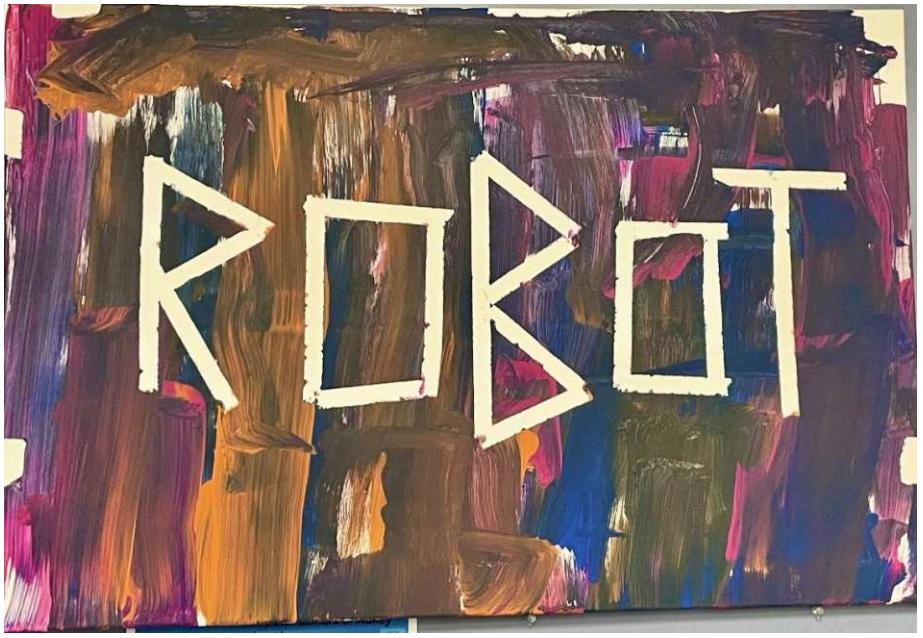
The SAR robotics team took a trip with all of our squads to the Spuyten Duyvil Library to present the three robots to children and adults there. We showcased the differences and abilities of all of our robots, wowing the crowd with airplane launches and explanations of all the tiny parts that work together to make our robot function. Many important local officials, including the New York City Mayor, attended this event, and we had the privilege of talking to them and telling them about FTC's robotics competitions.





Picasso's Painting:

For our school's annual night of art and music, we wanted to display to the crowds of students, parents, and teachers the fascinating additions technology can bring to artistic fields. We set up our robot front and center in the school's lobby along with its own paint colors, brush, and easel. Then the robot was controlled to splash the easel with a variety of colors by dipping its brush into the paint and then brushing it onto the canvas. At the end, we had a grand reveal where we peeled off the pre-placed tape on the robot to display the word "ROBOT" spelled out under all of the paint!

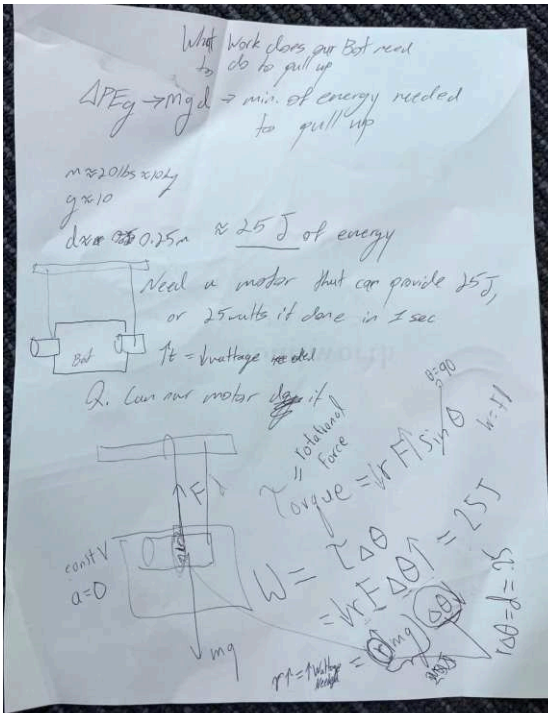


Girls Who Code Initiative:

Our team recognizes the challenges that girls face in fields such as robotics, and wanted to address this by offering tools to assist girls in learning skills such as coding earlier on. We created a digital classroom on CSAwesome, an online interactive coding textbook that teaches Java, and signed up girls who were interested from other SAR squads and the Girls Who Code club. This access allows them to learn Java at their own pace with the help of our team members and the thorough digital textbook (the same one is used in AP Computer Science).

Meeting with Physicist Ron Zamir:

We met with Mr. Zamir to figure out how on a physics level our robot would have to perform hanging on the trust and exactly how much energy would we need and what work would our robot have to do in order for the bot to do a pull up. Alongside Mr. Zamir we came to the decision that by latching onto the trust and applying rotational force we can use our motors to apply approximately 25 joules of torque and pull ourselves up. This is our next goal that we plan to achieve once we qualify.



Meeting with MIT Engineering Student Eden Hen

We met with Eden Hen, an engineering student at MIT and the previous year's SAR robotics captain. We talked to her about the FTC game this year and showed her our robot, and she spoke to us about the amazing work she is currently doing at MIT. Eden told us about how FTC was a great start in robotics for her and how even now at MIT she and her classmates use tech like april tags that they were all first introduced to through FTC. Eden also interned at the FTC Cornell Tech Campus over the summer and how it taught her the importance of networking and learning from other teams. She gave us the advice for our team growth to try reaching out to more teams to work with them and we decided to start making that a priority and planning ways to integrate it earlier for next year. It was inspiring for our team to see how the teamwork and knowledge that we gained from building for FTC can be applied to our future, and also important for us to realize new strategies that we should implement to better our team for next year!

PJ Jelen, Binghamton Engineering, University Rover Challenge

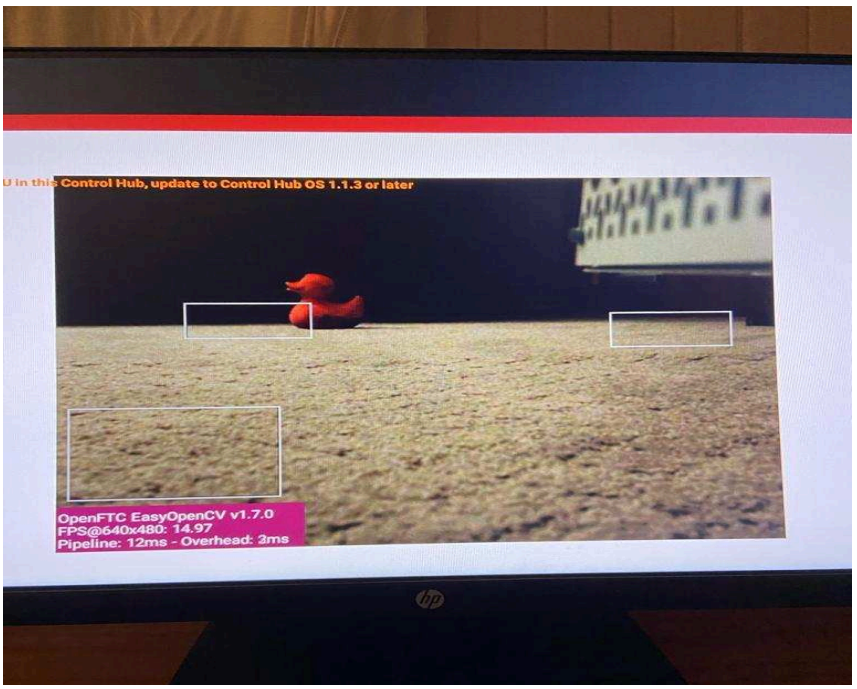
We met with PJ Jelen, who is currently working on his university's team for the Rover Challenge. He helps to build the Space Rover so that it can collect samples of the ground that can be tested to see if the location supports life. In our conversation explaining the FTC competition goal this year, we discovered many similarities and benefits between his current project and his time on the SAR robotics team. On his team, PJ also has to utilize autonomous coding and image recognition. He mentioned how being on the robotics team provided him with skills and experiences that are very helpful to him now, for example, the skills to think outside the box and problem-solve. It was exciting for our team to see how the skills we are learning now can be transferred in the future to something much bigger. He left us with some tips to constantly question our robot to see how we can improve it, and to not be afraid to mess up - failure is the first step to success.



VII: OBSTACLES

Autonomous Image Recognition:

When setting up the autonomous image recognition for our game prop, the robot had difficulty figuring out where to look for the prop to identify it. To fix this, we plugged the control hub into a monitor - this allowed us to see what the camera sees so that we could set up the correct coordinate searches and fix the image recognition!



Sustainability:

We prioritized it as a team to attempt to build our robot by reusing as many pieces as possible from previous years and holding back from purchasing new pieces. This relates to our team's values of sustainability and problem-solving through tough situations. This plan caused us to have to make some creative substitutions while building our robot. We utilized 3D printing to print parts that we did not already own, such as brackets, and sawed extrusions to create specific angled ends that we didn't already have. We had to think carefully about which parts to 3D print, however, because we realized later on that the weaker material caused stripping to easily occur. Therefore we swapped out plastic pieces with more pressure on them with metal.